

## LAB RECORD

### Password Strength Testing using Python

Write a Python program that accepts a password from the user and checks whether it is weak, medium, or strong based on the following rules:

- Minimum 8 characters
- Must contain at least one digit
- Must contain at least one uppercase letter
- Must contain at least one lowercase letter
- Must contain at least one special character

**Program: (File name: password\_checker.py)**

```
import re
```

```
def check_password_strength(password):
```

```
    if len(password) < 8:
```

```
        return "Weak: Password must be at least 8 characters long."
```

```
    if not any(char.isdigit() for char in password):
```

```
        return "Weak: Password must include at least one number."
```

```
    if not any(char.isupper() for char in password):
```

```
        return "Weak: Password must include at least one uppercase letter."
```

```
    if not any(char.islower() for char in password):
```

```
        return "Weak: Password must include at least one lowercase letter."
```

```
    if not re.search(r'[!@#$%^&*().,?":{}|<>]', password):
```

```
    return "Medium: Add special characters to make your password stronger."
```

```
    return "Strong: Your password is secure!"
```

```
def password_checker():
```

```
    print("Welcome to the Password Strength Checker!")
```

```
    while True:
```

```
        password = input("\nEnter your password (or type 'exit' to quit): ")
```

```
        if password.lower() == "exit":
```

```
            print("Thank you for using the Password Strength Checker! Goodbye!")
```

```
            break
```

```
        result = check_password_strength(password)
```

```
        print(result)
```

```
if __name__ == "__main__":
```

```
    password_checker()
```

### **Execution Steps:-**

1. Open Command Prompt or VS Code Terminal.
2. Navigate to the folder where password\_checker.py is saved
3. Run the program: python password\_checker.py

### **The program will display:**

Welcome to the Password Strength Checker!

Enter your password (or type 'exit' to quit):

4. Type a password and press Enter.
5. The program will display whether the password is:
  - Weak
  - Medium
  - Strong
6. To stop the program, type: exit

### **Sample Input & Output:**

#### **Input:**

Enter your password: abc123

#### **Output:**

Weak: Password must be at least 8 characters long.

#### **Input:**

Enter your password: Abcdef12

#### **Output:**

Medium: Add special characters to make your password stronger.

#### **Input:**

Enter your password: Abc@1234

#### **Output:**

Strong: Your password is secure!

## **Experiment: Analyzing Phishing Emails**

To analyze a suspicious email using EML Analyzer and VirusTotal and identify whether it is phishing based on technical indicators.

### **Tools Used:**

- Online EML Analyzer
- VirusTotal
- Sample file: 2020-05-05-phishing-email-example-01.eml

### **Questions to Answer:**

1. What is the full email address of the sender?
2. What domain is used to send this email?
3. What is the sender's IP address from the header?
4. Is the sender IP blacklisted?
5. What is the result of SPF authentication?
6. What is one suspicious URL found in the email body?

### **Part A: EML Analyzer Results**

The file 2020-05-05-phishing-email-example-01.eml was uploaded to the EML Analyzer.

EML Analyzer Results:

#### **Key Observations:**

- Subject: Warning: Final Notice
- From: malware-traffic-analysis.net Support [sues@nnwifi.com](mailto:sues@nnwifi.com)
- To: [brad@malware-traffic-analysis.net](mailto:brad@malware-traffic-analysis.net)
- Content Type: text/html
- Message-ID: Missing

#### **Header Analysis:**

- Sender IP: 94.100.31.27
- Reverse DNS: 94-100-31-27.static.hvvc.us
- Mail Server: mail.nnwifi.com

**Authentication:**

- SPF: Failed
- DKIM: Not signed
- DMARC: Not aligned

**Part B: VirusTotal Analysis**

The sender IP 94.100.31.27 was checked on VirusTotal.

**VirusTotal Result:**

- Detection Ratio: 1 / 93 vendors flagged as malicious
- Location: Netherlands
- ASN: AS29802 (HVC-AS)

This means the IP is not widely blacklisted but has suspicious reputation.

**Suspicious Link Identified**

From EML Analyzer, the following URL was extracted:

<https://servervirto.com.co/ed/trn/update?email=brad@malware-traffic-analysis.net>

Reasons it is suspicious:

- Does not match sender domain (nnwifi.com)
- Uses foreign domain (.com.co)
- Requests confirmation of ownership (credential harvesting pattern)

**Answers to Given Questions**

1. Full sender email address: sues@nnwifi.com
2. Domain used to send the email: nnwifi.com
3. Sender's IP address: 94.100.31.27
4. Is sender IP blacklisted? :Yes (1/93 vendors flagged it as malicious in VirusTotal)
5. SPF authentication result: Fail

6. One suspicious URL in email body:

<https://servervirto.com.co/ed/trn/update?email=brad@malware-traffic-analysis.net>

### **Conclusion:-**

Phishing indicators found:

- Urgent subject: "Warning: Final Notice"
- Fake display name pretending to be malware-traffic-analysis.net
- Actual sender domain is unrelated (nnwifi.com)
- SPF authentication failed
- Message-ID missing
- HTML-only email
- Suspicious external link
- IP partially flagged by VirusTotal

### **Final Verdict:-**

This email is classified as PHISHING.